

Binary Logit

Hypothesis Testing: Logistic Regression

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Where We Left Off

Multivariate regression is much more common than bivariate regression in publications.

Partial effects are the individual relationships (between each IV and the DV).

Control variables help us account for relationships in the background and pick up some “noise” (error) in the data.

Assumptions of OLS:

1. Linearity
2. Representative sampling
3. No multicollinearity (or perfect collinearity)
4. No heteroskedasticity
5. Independence (no autocorrelation)
6. Normally distributed errors, around 0.

Interpreting Categorical Variables

Interpreting regression models should be done systematically.

For *linear* regression, we need to consider the following:

- With continuous IVs, we can say that a one-unit in change in X leads to a (β coefficient) change in the DV.
- With categorical variables, this doesn't make logical sense!
- Dichotomous IV: we can say "As X moves from 0 to 1, it leads to..."
- Ordinal IV: Sometimes we can loosely interpret the results (when the ordinal measure is like an index or is big enough), but that's not always to best plan...
- Nominal IV: no ranked order means no clear linear pattern.

The problem is that ordinal and nominal variables do not always behave linearly!

Dummy It Out!

The solution is to make “dummy” variables from our nominal and ordinal variables.

“**Dummy**” variables are dichotomous level variables that we create from other data to represent a specific value (or lack of that value).

- Instead of a **nominal** variable for type of car (1: Toyota; 2: Ford; 3: Honda), we can make three dummy variables: Toyota or not, Ford or not, Honda or not.
- We can do the same thing with **ordinal** variables like support for a policy (1: Support strongly; 2: Support somewhat; 3: Neither support nor oppose; 4: Oppose somewhat; 5: Oppose Strongly): Support (strongly or somewhat) or not, etc.
- Even **continuous** variables can be used for a dummy: is it within a certain range or not, is it this specific value (like a year representing the start of Covid) or not, etc.

Logistic Regression: Logit

Logistic “regression” behaves differently than OLS regression.

- Logistic regression (logit) requires a *dichotomous* DV.
- Logistic regression helps us *predict* outcomes (DV) based on inputs (IV).

With logit, we are testing how likely an outcome is compared to what we’d expect if there were no relationship.

- H_0 : the predictor (IV) does not influence the likelihood of the DV ($\beta_1 = 0$)
- H_1 : the predictor (IV) influences the likelihood of the DV ($\beta_1 \neq 0$)

But interpretation is different for logit than it is for OLS:

- Logit doesn’t produce a line with a constant slope!
- The β coefficients/estimates mean something different: **log odds**.
- Because log odds are not directly interpretable, we usually convert them to **odds ratios** instead for a cleaner analysis.

The “logistic” part

OLS regression tries to fit a *line* to the data.

$$Y_i = \alpha + \beta_i X_i + \epsilon_i$$

But when there are only two outcomes, a straight line doesn't give us much information (and introduces a lot of error).

Logistic regression tries to fit a *logistic function* instead (an S-shaped curve).

$$P(Y_i = 1|X) = \frac{1}{1 + e^{-(\alpha + \beta_i X_i)}}$$

Maximum Likelihood Estimation (MLE)

Whereas linear regression relies on OLS, logit relies on MLE.

- We want to estimate the likelihood of being in the higher category of the DV (1 instead of 0).
- We want to find the S-shaped curve that *best* describes the data.

The maximum likelihood process models the “log odds” of the DV outcome happening.

- We start with probabilities of being a 1 on the DV.
- But the probabilities are nonlinear—they form an S-shaped curve of likelihood using the scary equation before.
- Because it's an S-shaped curve, the slope is not constant across a predictor, which means we transform them to log odds to make them linear for modeling.

The best logistic model will have the highest log odds (maximizing likelihood of $DV=1$).

Interpreting Logistic Regression Output

Recall the three big things to look for in *any* regression:

1. The significance of the model
2. The relationships between the IV(s) and DV.
3. The “goodness of fit” of the model

The “**goodness of fit**” is a metric for how well the model predicts the DV.

In *logistic* regression, we look at the “Pseudo R-Squared” values

- Pseudo R-Squared is not a great measure, but we sometimes still use it to compare across the other models we run.
- 0.0 to 0.1: weak
- 0.1 to 0.3: moderate
- 0.3 to 0.5: pretty strong
- 0.5 to 1.0: very strong [and highly unlikely!] (like, something might be wrong!)

Improving Output Interpretation

Logit partial effects:

- Each relationship between each IV and the DV.
- Log odds are hard to conceive of and interpret directly.
- So, we interpret them vaguely: the IV “increases/decreases the likelihood of the DV outcome...”

Or, we can use odds ratios instead

- We transform the log odds coefficients (β) to odds ratios: e^{β}
- Odds don't go negative, so 1.0 is the baseline: > 1.0 is an increase in odds, < 1.0 is a decrease.
- We can even be specific: 1.75 reflects a 75% increase in odds; 0.6 represents a 40% decrease in odds.

Better “goodness of fit”

- Beyond the somewhat confusing Pseudo R-squared, we can also just look at a classification table...
- Looking along the diagonal, we can see how many observations were correctly predicted and calculate the percentage of the DV the model explains.

Classification Tables

Classification Table			
Observed Outcomes	Predicted Outcomes		
	True	False	Total
True	6	21	27
False	7	135	142
Total	13	156	169

Look at the diagonal (\\) to draw conclusions:

- How many cases are both predict to be True and actually observed to be True?
- How many cases are both predict to be False and actually observed to be False?
- Add them up and divide by the total cases: $6 + 135 = 141/169 = 0.8343195$
- The model explains 83.4% of the variation in the DV!